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Natural Language Processing Applications



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Session 7: Privacy and Ethics in NLP Applications

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Agenda



01 Introduction to Privacy & Ethics in NLP

02 Data Issues — Collection, Consent & Bias

03 Model Issues — Fairness & Explainability

04 Deployment Issues — Safety & Accountability

05 Real-World Use Case Examples

06 Tools, Techniques & Production-Ready Setup

Why Privacy & Ethics Matter in NLP



Massive Data Usage

NLP systems ingest emails, chats, medical records — highly personal data



Re-identification Risk

Models can memorize and leak PII from training data



Bias & Discrimination

Biased training data leads to discriminatory outputs at scale



Regulations

GDPR, DPDP (India), CCPA (USA), EU AI Act impose strict obligations on AI deployments

82%

of users worry about AI privacy

\$20M

max GDPR fine per violation

4%

of global turnover penalty

Data Issues



Consent & Collection

- Scraping public data without explicit consent
- No opt-out mechanisms for data subjects
- Unclear data retention policies
- Lack of purpose limitation



Dataset Bias

- Underrepresented minorities in corpora
- Historical bias baked into text
- Label bias from annotators
- Geographic & language imbalance



PII & Leakage

Names, emails, SSNs in training data

Model memorization of rare examples

Inference attacks on embeddings

No data provenance tracking



Model Issues

Demographic Parity Violations

Sentiment models score identical text differently based on inferred gender, race, or religion

Lack of Explainability

Black-box transformers provide no rationale — impossible to audit for compliance (GDPR Art. 22)

Training Data Memorization

Large LLMs can regurgitate verbatim PII seen during training via extraction attacks

Calibration & Hallucination

Overconfident predictions mislead decision-makers; hallucinated facts cause downstream harm

Key Fairness Metrics

Demographic Parity

Equal positive rates across groups

Equal Opportunity

Equal true positive rates

Equalized Odds

Equal TPR & FPR across groups

Counterfactual

Prediction unchanged if group changes

Individual Fairness

Similar people treated similarly

Deployment Issues



No Output Filtering

Deployed models may generate harmful, hateful, or toxic outputs without guardrails



Lack of Monitoring

Absence of drift detection, fairness dashboards, or anomaly alerting post-deployment



Audit Trail Gaps

No logging of model decisions — unable to explain outcomes during regulatory audits



Access Control Failures

Broad API access without authentication; sensitive inference exposed to unauthorized users



Adversarial Attacks

Prompt injection, jailbreaks, and adversarial inputs bypass safety mechanisms



Vendor Lock-in Risk

Closed-source models offer no visibility into data handling, updates, or bias corrections



Tools & Techniques

Privacy & PII

- **Microsoft Presidio** PII detection & anonymization in text
- **spaCy + custom NER** Named entity masking pipeline
- **Google DLP API** Automated PII scanning at scale
- **Differential Privacy (DP-SGD)** Training with privacy guarantees

Explainability

- **SHAP** Shapley value-based feature importance
- **LIME** Local model-agnostic explanations
- **Captum (PyTorch)** Gradient-based interpretability
- **BertViz** Attention visualization for transformers

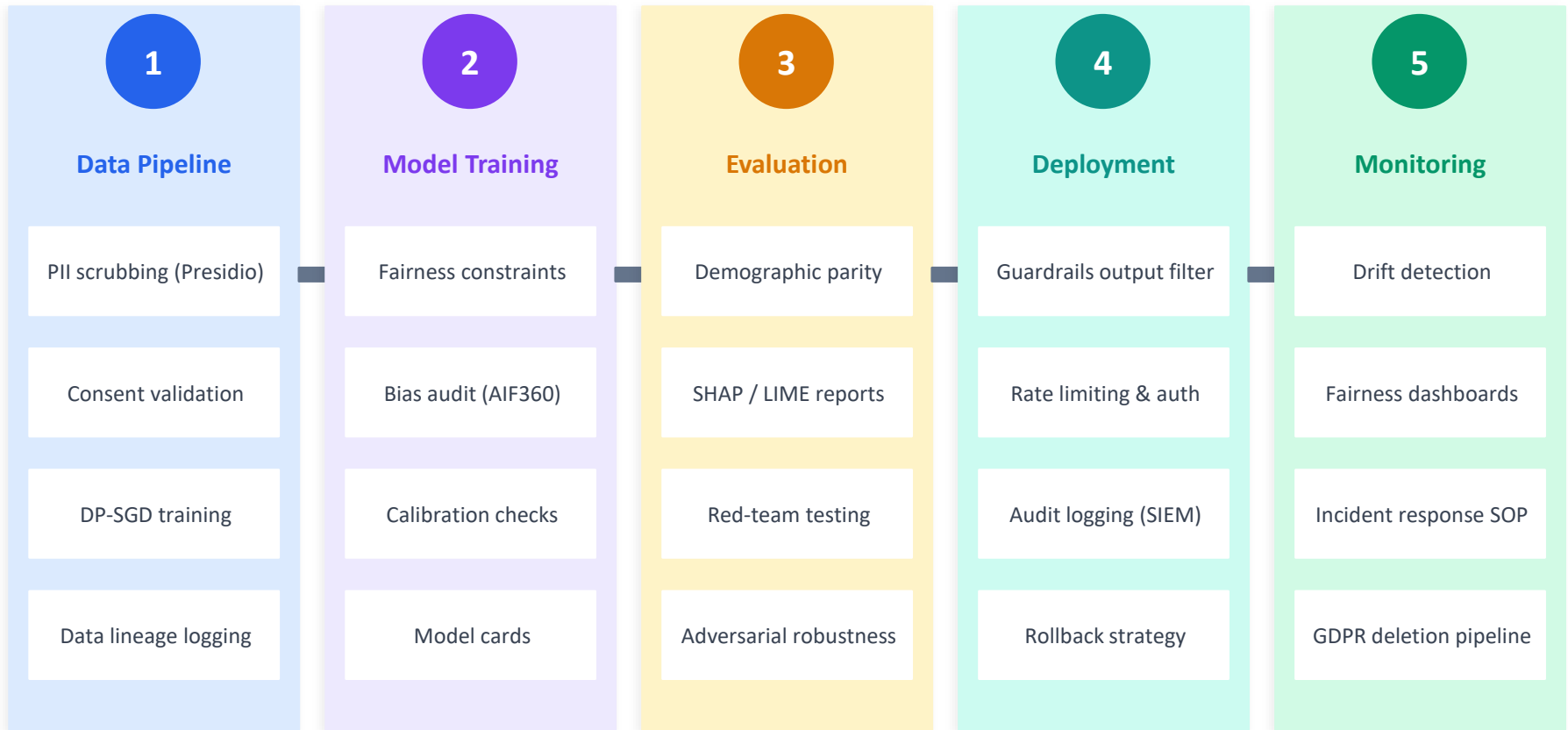
Fairness & Bias

- **IBM AI Fairness 360** Bias detection & mitigation toolkit
- **Fairlearn (Microsoft)** Fairness constraints in model training
- **What-If Tool (Google)** Visual model fairness exploration
- **Aequitas** Audit tool for ML pipelines

Safety & Red-Teaming

- **Garak** LLM vulnerability scanning
- **Guardrails AI** Output validation & constraint enforcement
- **LangKit / NeMo Guardrails** Toxicity & jailbreak detection
- **Rebuff** Prompt injection detection

Production-Ready Privacy & Ethics Setup





Ethics & Privacy Governance Checklist

Data

- ✓ Legal basis established for all training data
- ✓ PII identified and anonymized or removed
- ✓ Data retention & deletion policy documented
- ✓ Provenance and lineage tracked throughout pipeline

Model

- ✓ Fairness metrics evaluated across demographic groups
- ✓ Explainability report generated (SHAP/LIME)
- ✓ Model card published with limitations & risks
- ✓ Differential privacy applied where feasible

Deployment

- ✓ Output safety filters and guardrails in place
- ✓ Access controls and authentication enforced
- ✓ All decisions logged with timestamp & model version
- ✓ Incident response playbook documented

Ongoing

- ✓ Automated drift & fairness monitoring running
- ✓ Regular red-team / adversarial testing scheduled
- ✓ GDPR right-to-erasure pipeline implemented
- ✓ Ethics review board approval obtained



Privacy and Ethics in NLP Applications

- Personal and sensitive text data can be embedded in corpora, logs, and prompts.
- Fairness concerns arise when demographic, dialect, or domain representation is uneven.
- Ethical NLP requires consent awareness, traceability, transparency, and human accountability.
- Evaluation should extend beyond accuracy to include harm, bias, and misuse scenarios.

Connect ethics to the full NLP Application lifecycle, not only model training.

Data Issues

- Noisy, duplicated, or stale training data can reduce generalization and trustworthiness.
 - Annotation guidelines may encode human subjectivity or social stereotypes.
 - Data imbalance can suppress minority dialects, languages, or rare but critical intents.
 - Data governance must address provenance, retention, consent, and data minimization.
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Model Issues

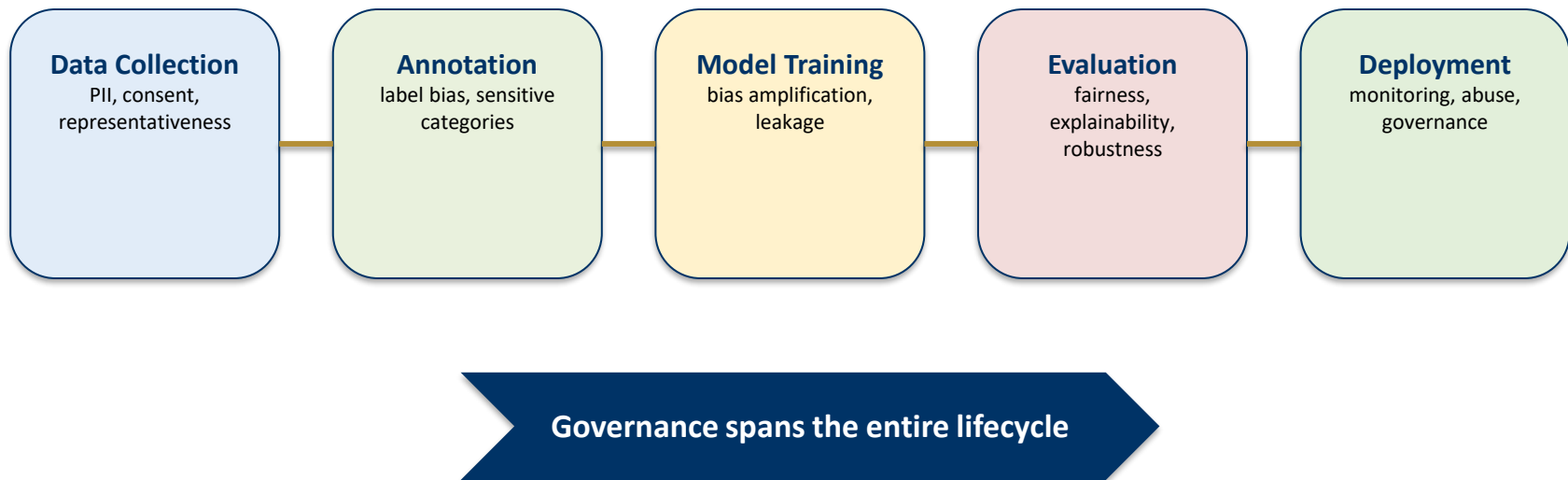
- Models can amplify patterns present in data, including harmful stereotypes.
 - Large models may hallucinate fluent but unsupported answers.
 - Explainability remains limited for many deep neural architectures.
 - Model behavior can shift across domains, languages, and adversarial inputs.
-



Deployment Issues

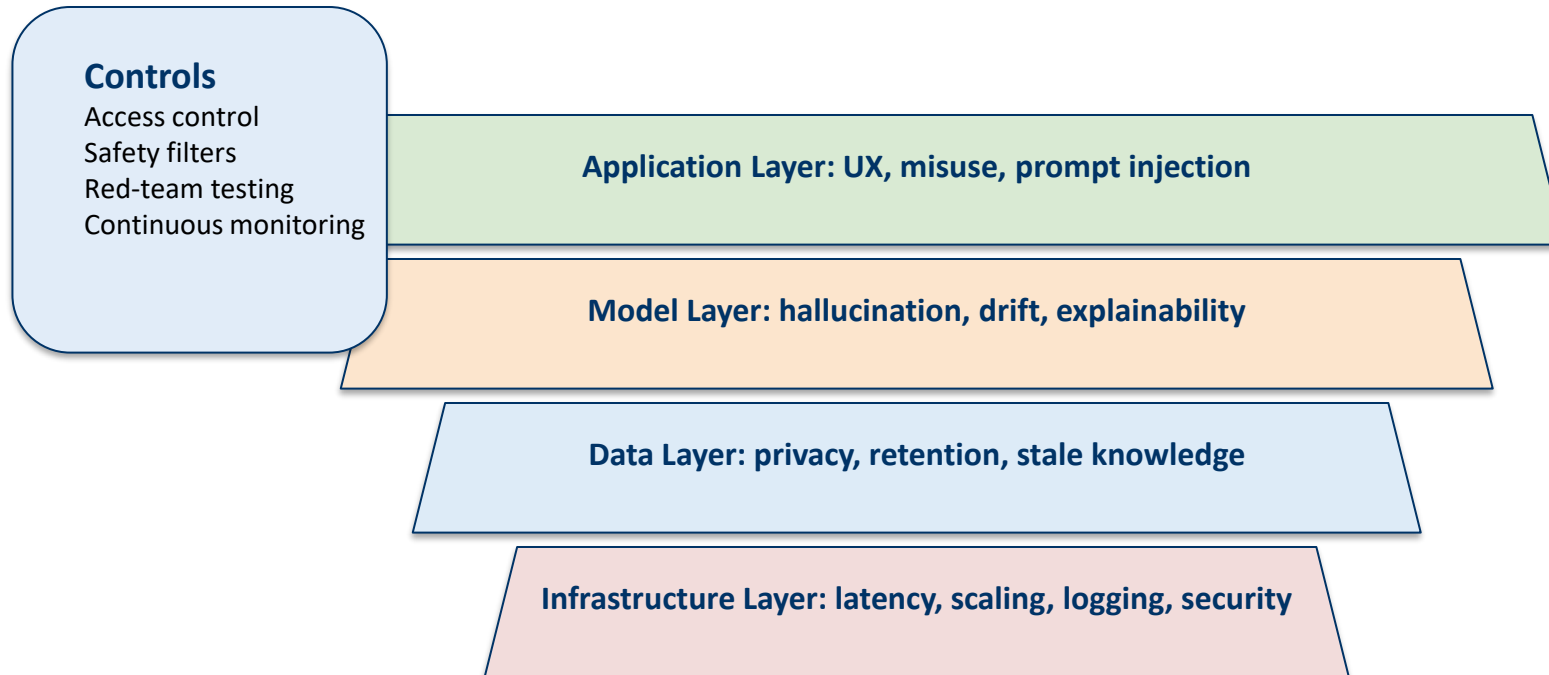
- Latency, scalability, and cost constraints affect user experience and operational viability.
 - Monitoring is essential for drift, abuse, privacy leakage, and safety regressions.
 - Fallback design and human escalation are necessary in high-stakes workflows.
 - Policies for logging, access, and red-team testing must be part of deployment governance.
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Diagram: Ethical Risk Lifecycle in NLP Applications



Bad Data → Biased Labels → Biased Model → Unfair Decisions → Harm

Diagram: Deployment Risk Stack for NLP Systems



Risk Stack in NLP Systems



- **Infrastructure Layer (scaling, security, logging)**
 - Example: OpenAI / ChatGPT outages & rate limits
 - High demand leads to latency issues and degraded user experience
 - Scalability failures directly impact usability and trust
- **Data Layer (privacy, retention, stale knowledge)**
 - Example: Samsung data leak via ChatGPT (2023)
 - Employees pasted sensitive code into chatbot
 - unintended data exposure
 - Poor data handling policies → privacy violations
- **Model Layer (hallucination, drift, explainability)**
 - Example: Google Bard factual errors during demo
 - Incorrect astronomical fact caused credibility issues
 - Hallucinations can damage trust, especially in high-stakes domains
- **Application Layer (misuse, prompt injection)**
 - Example: Prompt injection attacks on LLM apps
 - Malicious inputs bypass safeguards and extract hidden instructions
 - User-facing systems are vulnerable to adversarial misuse

Discussion

- 1 If a model is accurate overall but fails disproportionately on one user group, is deployment acceptable?
 - 2 What should be logged for debugging without violating privacy or retaining excessive personal data?
 - 3 Who is accountable when an NLP assistant gives harmful advice in a high-stakes domain?
 - 4 Should organizations prefer smaller, auditable models when explainability is more important than raw accuracy?
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Discussions:

Is deployment acceptable



- “Deploy the model (accuracy is acceptable)”
- “Do NOT deploy (fairness issues)”



Discussions: What to Log?



What to log without violating privacy?



Discussions: What to Log?

- Safe to log:
 - Model outputs
 - Timestamp
 - Model version
 - Error types / system metrics
- Avoid logging:
 - Raw user inputs containing PII
 - Full conversations (unless anonymized)
 - Sensitive personal identifier

- Key idea:

Log for debugging, but minimize personal data exposure



Discussions: Accountability

Responsibility for harmful outputs

Ask:

- Who is responsible?
 - Developer
 - Company
 - User
-



Discussions: Accountability

Primary accountability: Organization deploying the system

Shared responsibility:

- Developers → design & safeguards
- Company → deployment & monitoring
- Regulators → compliance enforcement

- Key idea:

Accountability is distributed but led by the deploying entity



Discussion:

Smaller vs large models?

Prefer explainable models when:

- Healthcare
- Finance
- Legal decision-making

Prefer large models when:

- Low-risk applications (chatbots, recommendations)

Key idea:

- Tradeoff between accuracy vs interpretability depends on risk level
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Activity: Model Choice Scenario

- Scenario:
 - Option A: Black-box LLM (high accuracy)
 - Option B: Smaller interpretable model

 - Task: Choose for-
 - Healthcare NLP system
 - Chatbot for customer support

 - Expected Insight:
 - High-stakes → explainability preferred
 - Low-stakes → accuracy acceptable
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Activity: Accountability mapping

Scenario:

An NLP healthcare chatbot gives incorrect medical advice, causing harm to a user.

Task:

1. Who should be responsible?

- Developer
- Company
- User
- Regulator



Case study: Chatbot data leaks

Case: A chatbot trained on customer data leaks sensitive user information.

Task:

1. Which stage caused the issue? (Data / Model / Deployment)
 2. What risks are involved? (privacy, bias, deployment, etc.)
 3. Suggest fixes
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Case study: Chatbot data leaks

Stage →

Data Collection + Training

Risk → PII leakage / memorization

Fix →

- PII scrubbing (Presidio)
- Differential privacy

Monitoring →

Output filters + red-teaming



Real-World Use Case Examples

Amazon Rekognition / NLP Hiring Tool

Data & Model
Bias

Amazon's ML hiring tool trained on 10 years of male-dominated resumes — systematically downgraded applications from women

- Training data must reflect desired future parity, not historical bias

GPT-3 / ChatGPT Memorization

PII Leakage

Researchers extracted verbatim personal data (emails, phone numbers) from GPT-3 via targeted extraction prompts

- Differential privacy & PII scrubbing must be applied before training at scale

Clearview AI

Consent Violation

Scraped 3B+ facial images without consent for a facial recognition NLP pipeline; fined €20M in Italy, banned in EU

- Data sourcing must include legal basis — consent or legitimate interest — under GDPR

Twitter Sentiment API

Deployment
Fairness

Twitter's photo-cropping algorithm (CLIP) consistently cropped out Black faces in favor of white faces

- Continuous fairness audits across demographic groups are required post-deployment



Industry Case Deep Dive: Amazon Recruiting Tool

Core issue: Bias in model outcomes

What happened

The system reportedly learned patterns from historical hiring data rather than an unbiased notion of merit.

Past data can reflect organizational imbalance, which then appears as a predictive signal to the model.

The case is widely used to teach why accuracy alone is not a sufficient deployment criterion.

Teaching takeaways

Dataset history matters as much as model architecture.

Fairness review should be part of feature design, threshold setting, and pre-deployment sign-off.

Human review must challenge model outputs rather than merely approve them.

Industry Case Deep Dive: Google Translate and Gender Stereotypes

Core issue: Stereotypical outputs drive

What happened

Translation systems can mirror social stereotypes present in multilingual text corpora.

Occupation-related translations are often discussed in class because they reveal how bias appears in a familiar application.

The example helps separate linguistic ambiguity from social bias in training data.

Teaching takeaways

Evaluation must include representative prompt sets, not just random samples.

Bias testing should cover occupations, pronouns, dialect variation, and low-resource settings.

Mitigation can involve balanced examples, constrained decoding, or post-processing policies.

Summary



- Data stage:
 - Ensure PII removal and user consent
- Training stage:
 - Apply differential privacy to prevent memorization
- Model stage:
 - Check for bias and data leakage
- Deployment stage:
 - Use guardrails and access control
- Monitoring stage:
 - Perform continuous auditing and data deletion support

Privacy and ethics must be integrated across the entire NLP lifecycle, not just at one stage
